EMCH 574 – Vibration and Waves
Homework 02

Student J.C. Vaught
Due date TBD

Contents

1 Waves **3**
1.1 A.1 Wave Fundamentals . . . 3
1.2 A.2 Simple Wave Calculations . . . 7
1.3 A.3 Calculate Car Motion on a Wavy Road . . . 11
1.4 A.4 Standing Waves . . . 12
1.5 A.5 Musical Notes . . . 13
1.6 A.6 Musical Instruments . . . 14

2 Fluid Waves and Vibration **15**
2.1 B.1 Analyze the Wave Equation in Fluids . . . 15
2.2 B.2 Analyze and Solve the Wave Equation . . . 16
2.3 B.3 Analyze Air Vibration in a Tube Closed at Both Ends . . . 17
2.4 B.4 Calculate Air Vibration in a Tube Closed at Both Ends . . . 19
2.5 B.5 Calculate the Slide Whistle . . . 20
2.6 B.6 Calculate Sound Intensity and the dB Scale . . . 22

3 Extra Credit Challenge Problems **23**
3.1 C.1 Analyze Air Vibration in a Closed-Open Tube . . . 23
3.2 C.2 Analyze Air Vibration in an Open-Open Tube . . . 25
3.3 C.3 Calculate Air Vibration in a Tube with Various Boundary Conditions . . . 26

Assignment

- G signifies students that take this class for Graduate Credit; they must do all the items.
- UG students are not required to do the G work, but they may do it for bonus points.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

Notes

- Show your work to obtain partial credit if appropriate.
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement reference.
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga, and so on) reference.
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit.
- Normalized plots are plots with the y-axis scaled between -1 and $+1$. A plotted variable is normalized by dividing it by its maximum absolute value.

Notations and Abbreviations

- BVP = boundary value problem.
- EOM = equation of motion.
- FBD = free body diagram.
- FRF = frequency response function.
- IVP = initial value problem.
- LHS = left hand side.
- NLM = Newton's law of motion.
- ODE = ordinary differential equation.
- PDE = partial differential equation.
- RHS = right hand side.

1 Waves

A.1 Wave Fundamentals

(1) List at least five examples of waves met in everyday life.

(2) Explain in your own words the difference between disturbance waves and continuous waves.

(3) Explain using text and sketches the definition of wavespeed for a disturbance wave.

(4) Give at least three different equivalent expressions for a disturbance wave.

(5) A car pileup takes place on an interstate and propagates as a wave for several miles. What type of wave is this? Choose one of the following:

- **(a)** Continuous wave.
- **(b)** Disturbance wave.

(6) Explain using text and sketches the definition of wavespeed for a continuous harmonic wave.

(7) Explain using text and sketches the definition of wavelength and wavenumber for a continuous harmonic wave.

(8) Give at least three different equivalent expressions for a continuous harmonic wave.

(9) What do waves carry? Choose one of the following:

- **(a)** Matter.
- **(b)** Energy.
- **(c)** Both.
- **(d)** None of the above.

(10) Graduate work. Write about the particle-wave duality of light.

Solution to A.1

1. Everyday examples of waves. Sound traveling through air, ripples spreading across water, light from a lamp, a pulse moving along a rope, and seismic waves traveling through Earth are familiar examples. Radio and Wi-Fi signals are also electromagnetic waves.

$$\boxed{\begin{array}{c}\text{Sound, water ripples, light, rope pulses,}\\ \text{seismic waves, and radio signals.}\end{array}} \tag{A.1.1}$$

2. Disturbance waves and continuous waves. A disturbance wave begins with a localized change, such as a single pulse, and carries that profile through a medium. A continuous wave is produced when the source oscillates repeatedly, creating a sustained sequence of crests and troughs. The distinction concerns how the source moves over time; both types propagate through space.

$$\boxed{\begin{array}{c}\text{A disturbance is localized in time.}\\ \text{A continuous wave is sustained by repeated oscillation.}\end{array}} \tag{A.1.2}$$

3. Wavespeed of a disturbance. Wavespeed is the distance a recognizable point on the disturbance travels divided by the elapsed time. For the two profiles below, the pulse advances by Δx during Δt, so its speed is $c = \Delta x/\Delta t$. For a right-moving pulse, $u(x,t) = F(x-ct)$ keeps the same shape because a fixed feature satisfies $x - ct = \text{constant}$.

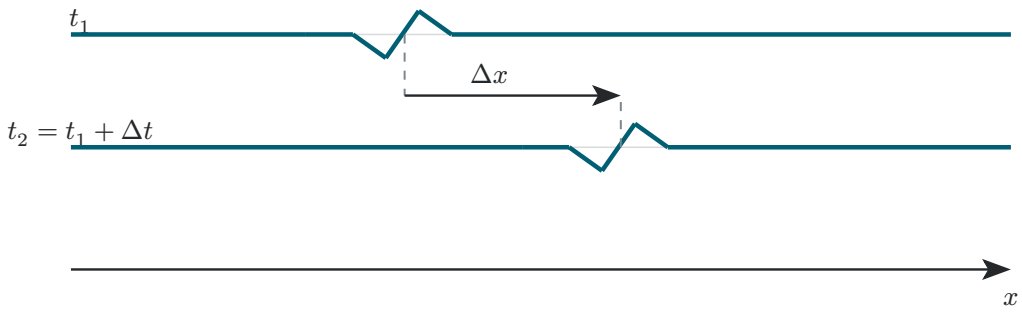


Figure 1: *A localized disturbance advances by Δx during Δt.*

$$\boxed{c = \frac{\Delta x}{\Delta t}} \tag{A.1.3}$$

4. Equivalent disturbance-wave expressions. For a right-traveling disturbance in a uniform medium with wavespeed c, let F describe its profile and let $\ell > 0$ be a reference length. Defining $G(\tau) = F(-c\tau)$ and $H(\xi) = F(\ell\xi)$ gives equivalent profiles written with a spatial, temporal, or dimensionless argument. Each expression therefore describes the same shape translating at speed c.

$$\boxed{u(x,t) = F(x-ct) = G\left(t - \frac{x}{c}\right) = H\left(\frac{x-ct}{\ell}\right)} \tag{A.1.4}$$

5. Interstate pileup. A car pileup begins locally and spreads as vehicles encounter the slowdown. It is therefore a disturbance wave.

$$\boxed{\text{Disturbance wave (choice b).}} \qquad \text{(A.1.5)}$$

6. Wavespeed of a continuous harmonic wave. A harmonic wave has a sinusoidal profile that repeats as it propagates. Wavespeed is the speed of a fixed phase point, such as a crest. If that crest moves by Δx during Δt, then $c = \Delta x/\Delta t$. For $u(x,t) = A\cos(\gamma x - \omega t + \phi)$, constant phase gives $\gamma x - \omega t + \phi = \text{constant}$ and hence $c = \omega/\gamma$.

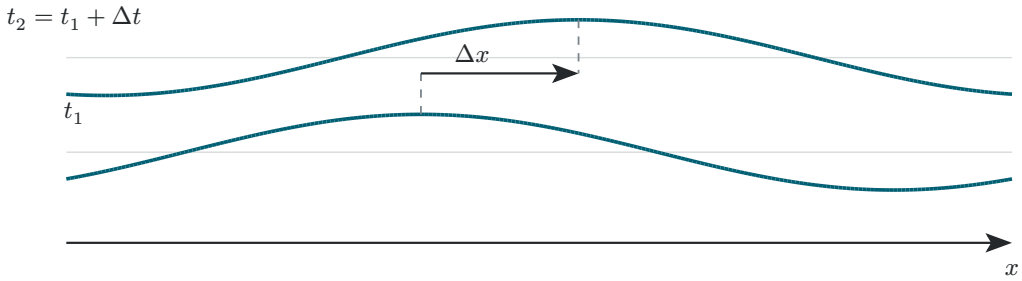


Figure 2: *A fixed phase point advances with the continuous harmonic profile.*

$$\boxed{c = \frac{\Delta x}{\Delta t} = \frac{\omega}{\gamma} = f\lambda} \qquad \text{(A.1.6)}$$

7. Wavelength and wavenumber. Wavelength λ is the spatial distance over which the harmonic profile completes one cycle, measured between adjacent points in the same phase, such as neighboring crests. Wavenumber γ is the phase change per unit distance. Since one spatial cycle advances the phase by 2π radians over distance λ, the relationship is $\gamma = 2\pi/\lambda$. Its SI unit is radians per meter, commonly written rad/m.

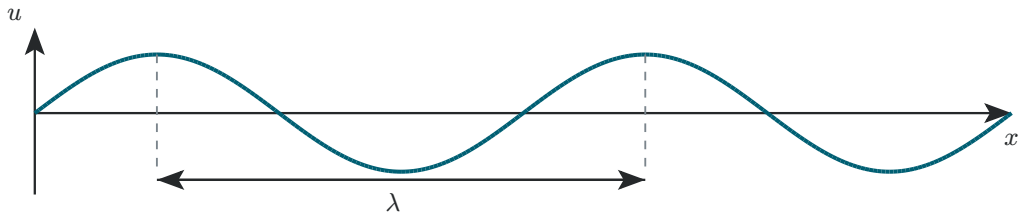


Figure 3: *The wavelength spans one complete spatial cycle between matching phase points.*

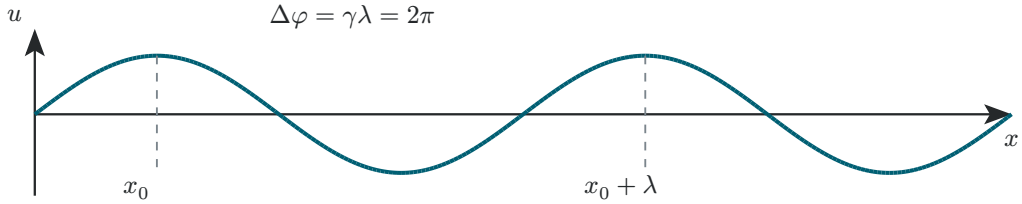


Figure 4: *Across one wavelength, the harmonic phase increases by 2π radians.*

$$\boxed{\gamma = \frac{2\pi}{\lambda}, \qquad \lambda = \frac{2\pi}{\gamma}} \tag{A.1.7}$$

8. Equivalent continuous harmonic-wave expressions. A traveling harmonic wave can be written as a cosine with phase, a sine with an adjusted phase, or as the real part of a complex exponential. These forms describe the same oscillatory motion with suitable choices of amplitude and phase. Here A is the amplitude, ω is angular frequency, γ is wavenumber, and ϕ is phase.

$$\boxed{\begin{aligned} u(x,t) &= A\cos(\gamma x - \omega t + \phi) \\ &= A\sin(\gamma x - \omega t + \phi + \tfrac{\pi}{2}) \\ &= \mathrm{Re}\Big\{ A e^{i(\gamma x - \omega t + \phi)} \Big\} \end{aligned}} \tag{A.1.8}$$

9. What waves carry. A wave transports energy through space or through a medium. In a mechanical wave, particles of the medium oscillate around their equilibrium positions; the wave does not carry the medium along as a net flow of matter.

$$\boxed{\text{Energy (choice b).}} \tag{A.1.9}$$

10. Graduate work. Particle-wave duality of light. Light propagates and interferes as an electromagnetic wave, yet exchanges energy with matter in discrete photons. A photon of frequency f carries energy $E = hf$, where h is Planck's constant; for light in vacuum its momentum is $p = h/\lambda$ [3]. The photoelectric effect shows that energy transfer occurs in photon-sized packets, while interference and diffraction reveal wave behavior. These observations reveal complementary aspects of light. Its wave behavior governs propagation and interference, while photon events describe localized exchanges with detectors and matter [2].

$$\boxed{E = hf, \qquad p = \frac{h}{\lambda}} \tag{A.1.10}$$

A.2 Simple Wave Calculations

Use the electromagnetic wavespeed $c_{\text{light}} = 300{,}000\,\text{km/s}$ and sound wavespeed $c_{\text{sound}} = 340\,\text{m/s}$.

Do the following.

(1) Calculate the frequency of visible light with typical wavelength $\lambda = 510\,\text{nm}$.

(2) Calculate the Hz frequency f of an AM radio wave with wavelength $\lambda = 550\,\text{m}$.

(3) Infrared radiation has a typical wavelength $\lambda = 10\,\mu\text{m}$. What is the typical frequency of infrared radiation?

(4) Calculate the wavelength λ of an FM radio wave with frequency $f = 89\,\text{MHz}$.

(5) The lightning flash is observed at 8:23:32 p.m., whereas the thunder is heard at 8:23:36 p.m. How far away is the lightning strike? Give your answer in kilometers and miles.

(6) One of Verizon's 5G mobile signals has frequency $f \approx 29\,\text{GHz}$. What is the corresponding wavelength?

(7) A microwave oven typically operates at $f = 2.5\,\text{GHz}$.

- **(a)** What is the wavelength λ?
- **(b)** What is the maximum permissible hole size D in the oven's door mesh, given that the largest hole in a Faraday cage must be at most $\lambda/2$?
- **(c)** Measure approximately the hole size in a microwave oven door mesh and discuss your results. You may search the web for various opinions on this matter.

(8) Graduate work. Given that the diameter of an electron is estimated to be $D = 2\,\text{pm}$, complete items (a) and (b).

- **(a)** Calculate the frequency of the electromagnetic wave that has a wavelength equal to the electron diameter.
- **(b)** In what spectrum range would this electromagnetic wave be? (options: infrared, visible, ultraviolet, X-ray, gamma ray).

Solution to A.2

In one period T, a wave crest travels one wavelength λ. Since $f = 1/T$, the wavespeed relation is

$$c = \frac{\lambda}{T} = \lambda f, \qquad f = \frac{c}{\lambda}, \qquad \lambda = \frac{c}{f}.$$

The supplied speeds in SI units are $c_{\text{light}} = 3.00 \times 10^8$ m/s and $c_{\text{sound}} = 340$ m/s. These values are used throughout; unit prefixes are converted before substitution.

For A.2.1, the visible-light wavelength is $\lambda = 510$ nm $= 510 \times 10^{-9}$ m. Therefore,

$$\boxed{f = \frac{3.00 \times 10^8}{510 \times 10^{-9}} \text{ Hz} = 5.88 \times 10^{14} \text{ Hz} = 588 \text{ THz}} \qquad \text{(A.2.1)}$$

For A.2.2, the AM wavelength is $\lambda = 550$ m. Using the same relation gives

$$\boxed{f = \frac{3.00 \times 10^8}{550} \text{ Hz} = 5.45 \times 10^5 \text{ Hz} = 545 \text{ kHz}} \qquad \text{(A.2.2)}$$

For A.2.3, the infrared wavelength is $\lambda = 10\ \mu$m $= 10 \times 10^{-6}$ m. Its frequency is

$$\boxed{f = \frac{3.00 \times 10^8}{10 \times 10^{-6}} \text{ Hz} = 3.00 \times 10^{13} \text{ Hz} = 30.0 \text{ THz}} \qquad \text{(A.2.3)}$$

The next item supplies frequency instead of wavelength. For A.2.4, $f = 89$ MHz $= 89 \times 10^6$ Hz, so

$$\boxed{\lambda = \frac{3.00 \times 10^8}{89 \times 10^6} \text{ m} = 3.37 \text{ m}} \qquad \text{(A.2.4)}$$

For A.2.5, the observed delay between the flash and thunder is $\Delta t = 36 - 32 = 4$ s. Taking the light and sound to originate at the same instant and travel the same distance d, their arrival-time difference is

$$\Delta t = \frac{d}{c_{\text{sound}}} - \frac{d}{c_{\text{light}}}.$$

Solving this equation for distance accounts for both travel times and gives

$$d = \frac{\Delta t}{1/c_{\text{sound}} - 1/c_{\text{light}}} = \frac{4}{1/340 - 1/(3.00 \times 10^8)} \text{ m}.$$

Using 1 km $= 1000$ m and the exact conversion 1 mi $= 1609.344$ m, the required results are

$$\boxed{d = 1.360 \text{ km} = 0.845 \text{ mi}} \qquad \text{(A.2.5)}$$

Because light travels much faster than sound, the usual approximation $d \simeq c_{\text{sound}}\Delta t = 340(4) = 1360$ m gives the same reported answers. The finite-light correction is only 1.541 mm.

For A.2.6, the supplied mobile-signal frequency is $f \approx 29$ GHz $= 29 \times 10^9$ Hz. Its wavelength is

$$\boxed{\lambda = \frac{3.00 \times 10^8}{29 \times 10^9}\ \text{m} = 10.34\ \text{mm}} \qquad \text{(A.2.6)}$$

The microwave oven in A.2.7 operates at $f = 2.5$ GHz $= 2.5 \times 10^9$ Hz. Applying the same wavespeed relation answers part (a),

$$\boxed{\lambda = \frac{3.00 \times 10^8}{2.5 \times 10^9}\ \text{m} = 0.1200\ \text{m} = 120.0\ \text{mm}} \qquad \text{(A.2.7.a)}$$

For part (b), the problem specifies $D \leq \lambda/2$. Consequently, the largest diameter allowed by that stated criterion is

$$\boxed{D_{\text{max}} = \frac{\lambda}{2} = \frac{0.1200}{2}\ \text{m} = 60.0\ \text{mm}} \qquad \text{(A.2.7.b)}$$

For A.2.7.c, we measured the mesh on our Oster small office microwave and found a hole diameter of roughly 1.2 mm and a center-to-center pitch of roughly 1.6 mm. Using the measured diameter gives

$$\boxed{D_{\text{mesh}} \approx 1.2\ \text{mm}, \qquad \frac{D_{\text{mesh}}}{\lambda} \approx 0.01000, \qquad \frac{D_{\text{max}}}{D_{\text{mesh}}} \approx 50.0} \qquad \text{(A.2.7.c)}$$

Thus, the measured hole is approximately one hundredth of the microwave wavelength and fifty times smaller than the classroom upper bound. Its opening is nevertheless much larger than a visible-light wavelength, so food remains visible through the perforated screen. The 60.0 mm result answers the supplied $\lambda/2$ criterion; it is not a microwave-door design specification.

For the graduate calculation in A.2.8.a, use the stipulated length $D = 2$ pm $= 2 \times 10^{-12}$ m as the electromagnetic wavelength. Then

$$\boxed{f = \frac{c_{\text{light}}}{D} = \frac{3.00 \times 10^8}{2 \times 10^{-12}}\ \text{Hz} = 1.500 \times 10^{20}\ \text{Hz} = 150.0\ \text{EHz}} \qquad \text{(A.2.8.a)}$$

This calculation uses the supplied length scale without establishing a physical electron diameter. The resulting wavelength is 0.00200 nm, below the 0.01 nm gamma-ray range described by NASA[1]. Therefore, the wavelength-based classification requested in A.2.8.b is

$$\boxed{\text{Gamma ray}} \qquad \text{(A.2.8.b)}$$

MATLAB code. *Script used to reproduce the numerical results in A.2.*

```
% A.2. Frequency, wavelength, lightning range, and microwave mesh.
cLight = 300000*1000;              % m/s, supplied light wavespeed
cSound = 340;                      % m/s, supplied sound wavespeed
mPerMile = 1609.344;               % m/mi, exact conversion
lambdaVisible = 510e-9;            % m
lambdaAM = 550;                    % m
lambdaIR = 10e-6;                  % m
fFM = 89e6; f5G = 29e9;            % Hz
fMicrowave = 2.5e9;                % Hz
flashSecond = 32; thunderSecond = 36;
Dmesh = 1.2e-3;                    % m, approximate measured hole diameter
meshPitch = 1.6e-3;                % m, approximate measured center pitch
DelectronGiven = 2e-12;            % m, stipulated length for item 8
disp(table(cLight,cSound,mPerMile));
disp(table(lambdaVisible,lambdaAM,lambdaIR,fFM,f5G,fMicrowave));
disp(table(flashSecond,thunderSecond,Dmesh,meshPitch,DelectronGiven));

% A.2.1--4 and A.2.6. Use c = lambda*f with SI inputs.
fVisible = cLight/lambdaVisible;
fAM = cLight/lambdaAM;
fIR = cLight/lambdaIR;
lambdaFM = cLight/fFM;
lambda5G = cLight/f5G;
disp(table(fVisible,fAM,fIR,lambdaFM,lambda5G, ...
    'VariableNames',{'visible_Hz','AM_Hz','infrared_Hz', ...
    'FM_m','mobile_m'}));

% A.2.5. Arrival delay equals sound travel time minus light time.
delay = thunderSecond-flashSecond;
distance = delay/(1/cSound-1/cLight);
distanceApprox = cSound*delay;
distanceKm = distance/1000;
distanceMiles = distance/mPerMile;
lightCorrectionMm = (distance-distanceApprox)*1000;
disp(table(delay,distanceKm,distanceMiles,lightCorrectionMm));

% A.2.7. The half-wavelength limit is the assignment's criterion.
lambdaMicrowave = cLight/fMicrowave;
Dmax = lambdaMicrowave/2;
meshToWavelength = Dmesh/lambdaMicrowave; % approximate measured ratio
boundToMesh = Dmax/Dmesh;          % approximate measured ratio
disp(table(lambdaMicrowave,Dmax,Dmesh,meshPitch, ...
    meshToWavelength,boundToMesh));

% A.2.8. Treat the given 2 pm length as the wavelength.
fElectronLength = cLight/DelectronGiven;
disp(table(fElectronLength));
```

A.3 Calculate Car Motion on a Wavy Road

Consider a car traveling on a wavy road. The road surface undulates with an amplitude $a = 2.5\,\text{in}$ and a wavelength $\lambda = 105\,\text{ft}$ that follows a cosine shape. The car is modeled as a spring-mass-damper (Figure 5) with natural frequency $f_n = 0.95\,\text{Hz}$ and damping ratio $\zeta = 22\%$. The maximum car speed is $v_{\text{max}} = 125\,\text{mph}$.

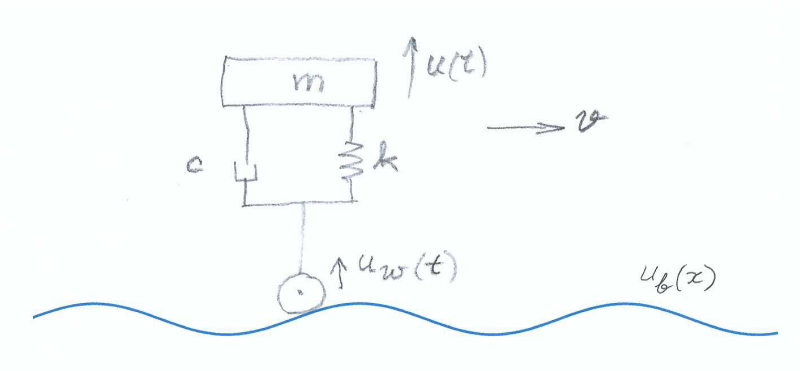


Figure 5: *Car riding on a wavy road.*

Do the following.

(1) Display input data.

(2) Calculate wavy road wavenumber γ.

(3) Plot the wavy road profile $u_b(x)$ for $N_\lambda = 2$ wavelengths.

(4) Calculate wavy-road frequency f_{max} for a maximum car speed of v_{max} and define the frequency range $f = \text{linspace}(0, f_{\text{max}}, N_f)$ with $N_f = 2000$.

(5) Plot $|\text{FRF}(f; f_n)|$. Use datatips to locate the peak response and the matched-frequency response.

(6) Calculate the matched-frequency response value $|\text{FRF}_n|$. Compare with the matched-frequency readings at item (5).

(7) Calculate the car speed v_n needed to make wavy-road frequency f match the car natural frequency f_n.

(8) Use the conversion $f = v/\lambda$ to plot $|\text{FRF}(v; f_n)|$. Use datatips on the plot to find:

- **(a)** Find v_n and $|\text{FRF}_n|$, and compare them with the results of items (6) and (7).
- **(b)** Find v_{cr} and $|\text{FRF}_{\text{max}}|$.
- **(c)** The limit speed v_{limit} for which the motion amplification is limited to $|\text{FRF}_{\text{lim}}| = 1.5$.

(9) Graduate work. Find the road wavelength λ_{new} that will make $v_n = 35\,\text{mph}$.

(10) Graduate work. Repeat items (5) through (8) for a car with worn-out shocks that provide only $\zeta = 10\%$, and compare the two cases.

A.4 Standing Waves

(1) Follow the notes to develop the standing-wave concept:

- **(a)** Write the complex-exponential expressions for two identical harmonic waves of amplitude $1/2$ propagating in opposite directions.
- **(b)** Add the two waves and write the resulting complex-exponential expression.
- **(c)** Use Euler's formula to arrive at a trigonometric expression.

(2) Sketch a standing wave and identify the nodes and antinodes (bows).

(3) Assume wavespeed $c = 340\,\mathrm{m/s}$ and frequency $f = 2.9\,\mathrm{kHz}$. Calculate:

- **(a)** Rad/sec frequency ω.
- **(b)** Wavenumber γ.
- **(c)** Wavelength λ.
- **(d)** Distance Δx between standing wave nodes.

(4) Graduate work. Explain intuitively the analogy between standing waves and the vibration of a taut string.

A.5 Musical Notes

(1) Find the Hz frequency of note C_5.

(2) Find what musical note corresponds to 246.94 Hz.

(3) What is the lowest frequency on the scale of equal temperament and what musical note does it correspond to?

(4) What is the commonly accepted frequency range of human hearing?

(5) What is the highest frequency on the scale of equal temperament and what musical note does it correspond to?

(6) Graduate work. Explain the scale of equal temperament by answering items (a)–(c).

- **(a)** What is the organizing principle of the scale of equal temperament.
- **(b)** How many equal-temperament entries are in an octave?
- **(c)** What is the ratio between two adjacent entries on the scale of equal temperament?

A.6 Musical Instruments

(1) List at least three methods to produce sound in a string instrument.

(2) List the two different methods to produce a sound in wind instruments.

(3) List string instruments that use a hammer hit to produce sound.

(4) List the three major categories of musical instruments. For each category, complete items (a) and (b).

- **(a)** Describe how the instrument works.
- **(b)** Explain what physical principles are used to produce sound.

(5) List at least five common string instruments.

2 Fluid Waves and Vibration

B.1 Analyze the Wave Equation in Fluids

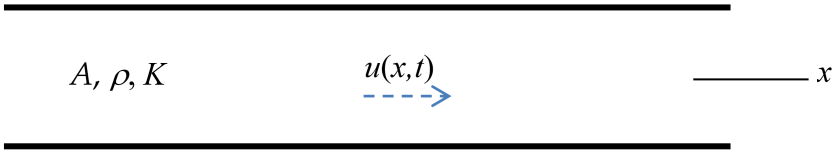


Figure 6: *Fluid waves in an infinite tube.*

Assume an infinite tube of cross section A containing fluid of density ρ and bulk modulus K (Figure 6). Develop the wave equation in fluids.

Do the following.

(1) Assume that the fluid in the tube undergoes time-varying axial displacement $u(x,t)$ under the influence of a pressure disturbance $p(x,t)$.

(2) Consider an infinitesimal element dx and draw the displacements and pressures on both of its sides.

(3) Draw FBD.

(4) Apply NLM and deduce relationship between pressure p, density ρ and acceleration $\ddot{u}$.

(5) Apply fluid-elasticity equation and express the pressure $p(x,t)$ in terms of displacement $u(x,t)$ and bulk modulus K.

(6) Substitute (5) into (4) and deduce the relationship between the second derivative with respect to space, u'', and the second derivative with respect to time, $\ddot{u}$.

(7) Define the wavespeed c.

(8) Substitute (7) into (6) and deduce the wave equation.

B.2 Analyze and Solve the Wave Equation

Recall the results of Problem B.1.

Do the following.

(1) Recall the wave equation.

(2) Assume separation of time and space behaviors into $U(x)$ and $T(t)$ and write $u(x,t)$ as a function of $U(x)$ and $T(t)$.

(3) Perform the appropriate differentiations with respect to space and time.

(4) Assume vibratory time behavior with angular frequency ω using a complex exponential formula for $T(t)$.

(5) Substitute (4) into (3) and write expressions for u'' and $\ddot{u}$.

(6) Substitute (5) into the wave equation (1) and express it only in terms of $U(x)$, c, ω.

(7) Define wavenumber γ.

(8) Substitute (7) into (6) and obtain a linear ordinary differential equation (ODE) in terms of $U(x)$, $U''(x)$, and γ.

(9) Assume exponential solution of the form $U(x) = e^{sx}$.

(10) Substitute (9) into (8) and obtain the characteristic equation.

(11) Solve the characteristic equation (10) and obtain the roots s_1, s_2.

(12) Substitute (11) into (9) and write the complete solution in terms of constants A_1, A_2 and complex exponentials of s_1, s_2.

(13) Add the vibratory time behavior from (4) to obtain the solution in terms of complex exponentials depending on both γ and ω.

(14) Use Euler's formulas to rewrite (13) in terms of new constants C_1 and C_2, trigonometric functions depending on γ, and a complex exponential depending on ω.

B.3 Analyze Air Vibration in a Tube Closed at Both Ends

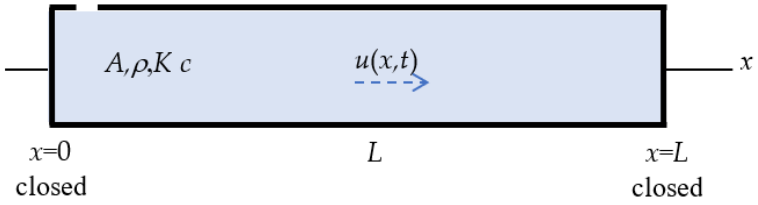


Figure 7: *Air vibration in a tube of finite length L closed at both ends.*

Assume a tube of finite length L and cross section A containing fluid of density ρ, bulk modulus K, and wavespeed c (Figure 7). The tube is closed at both ends. The air in the tube undergoes harmonic axial displacement $u(x,t)$ with frequency ω.

Do the following.

(1) Write the boundary conditions at the two ends of the tube in terms of displacement $u(x,t)$.

(2) Recall the harmonic solution of the wave equation consisting of the multiplication of two terms:

- **(a)** First term: a function $U(x)$ consisting of trigonometric functions depending on wavenumber γ, space variable x, and constants C_1, C_2.
- **(b)** Second term: a complex exponential depending on frequency ω and time variable t.

(3) Substitute (2) into (1) and write the boundary conditions at the two ends of the tube in terms of unknown constants C_1, C_2.

(4) Solve (3) for C_2 and obtain an equation for C_1.

(5) Discuss the trivial solution.

(6) Choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution.

(7) Introduce the notation $z = \gamma L$ into (6) and write the eigenvalue equation in terms of z.

(8) Solve the eigenvalue equation and write the general expression for the eigenvalues z_j, where $j = 1, 2, 3, \ldots$.

(9) Use (7) to write the general expression for the wavenumber γ in terms of j and length L.

(10) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c.

(11) Solve (10) for frequency ω in terms of wavenumber γ and wavespeed c.

(12) Use (9) and (11) to write the general expression of the natural frequency ω_j in terms of j, wavespeed c, and tube length L.

(13) Use (12) to write the general expression of the Hz frequency f in terms of j, wavespeed c, and tube length L.

(14) Explain the meaning of fundamental frequency and overtones.

(15) Explain the meaning of fundamental harmonic and higher harmonics.

(16) Recall the relationship between wavelength λ and wavenumber γ.

(17) Substitute (9) into (16) to write the general expression for the wavelength λ_j in terms of j and tube length L.

(18) Use (17) to explain which wavelength multiples could accommodate vibratory standing waves in a tube of length L closed at both ends.

(19) Recall $U(x)$ from (2)(a) and use (4) to eliminate C_2.

(20) Set $C_1 = 1$ in item (19) and write the general expression for the displacement modeshape $U_j(x)$ in terms of the wavenumber γ_j.

(21) Write the general expression of the pressure modeshapes.

B.4 Calculate Air Vibration in a Tube Closed at Both Ends

Consider a tube of length $L = 0.85\,\mathrm{m}$ closed at both ends. Take sound speed $c = 340\,\mathrm{m/s}$ and air density $\rho = 1.2\,\mathrm{kg/m^3}$. Use the results of Problem B.3 to calculate the modal vibration characteristics and modeshapes for $j = 1, \ldots, N$ with $N = 5$.

(1) Display input data.

(2) Calculate and display the modal vibration characteristics:

- **(a)** Eigenvalues z_j.
- **(b)** Wavenumbers γ_j.
- **(c)** Rad/s frequencies ω_j.
- **(d)** Hz frequencies f_j.

(3) Calculate and plot the displacement modeshapes.

(4) Discuss your results.

(5) Graduate work. Calculate and plot the normalized pressure modeshapes, and compare the result with item (3).

B.5 Calculate the Slide Whistle

Consider the slide whistle shown in Figure 8. The fully retracted whistle is shown in Figure 8(a), whereas the fully extended whistle is shown in Figure 8(b). The length of the residual cavity of the fully retracted whistle tube can be approximated to be $L_0 = 6\,\text{mm}$ (Figure 8(a)). The thickness of the piston inside the whistle can be approximated to be $L_p = 6\,\text{mm}$. The thickness of the end walls can be approximated to be $L_w = 6\,\text{mm}$. Assume sound speed in air to be $c = 340\,\text{m/s}$. The purpose of this problem is to calculate the fundamental frequency for various positions of the whistle slider



(a)



(b)

Figure 8: *Slide whistle. (a) Fully retracted, showing the piston end. (b) Fully extended. The ruler's major markings are in centimeters.*

Recall the theoretical results of Problem B.3.

Do the following.

(1) Display input data.

(2) Use Figure 8 to estimate the fully extended whistle length L_e.

(3) Use Figure 8 to estimate the fully contracted whistle length L_c.

(4) Use L_e, L_c to calculate the extension travel dL_I corresponding to fully extended slide whistle.

(5) Use dL_I and L_0 to calculate the cavity length L_I of a fully extended slide whistle.

(6) Estimate the fully extended frequency f_I in Hz and find the nearest musical note expressed as a letter name and octave number (e.g., G3 signifies note G in the third octave).

(7) Use L_e, L_c to calculate the extension travel dL_{II} corresponding to 40% extended slide whistle.

(8) Use dL_{II} and L_0 to calculate the cavity length L_{II} of a 40%-extended slide whistle.

(9) Calculate the 40%-extended frequency f_{II} in Hz and find the nearest musical note expressed as a letter name and an octave number.

(10) Comment on your work.

Graduate Work

Estimate the piston position required to obtain the musical note B7.

Do the following.

(11) Display the frequency f_{III} of the note B7.

(12) Calculate the cavity length L_{III} needed to generate frequency f_{III} of the note B7.

(13) Calculate the extension travel dL_{III} needed to generate frequency f_{III} of the note B7.

(14) Estimate what percentage of the piston must be extended to generate the note B7.

(15) Comment on your work.

B.6 Calculate Sound Intensity and the dB Scale

Take sound speed $c = 340\,\mathrm{m/s}$ and air density $\rho = 1.2\,\mathrm{kg/m^3}$.

Do the following.

(1) Display input data.

(2) Calculate air's acoustic impedance Z.

(3) Calculate the sound intensity I corresponding to a sound pressure $p = 55\,\mathrm{mPa}$.

(4) Give the values of the reference sound intensity I_0 and reference sound pressure p_0.

(5) Convert the sound pressure and sound intensity from item (3) to decibels.

(6) Verify that the sound pressure and sound intensity calculated at item (5) have the same dB value and explain why it is so.

(7) Calculate in physical units the sound intensity, sound pressure, and sound power from a loudspeaker with diameter $D = 1.55\,\mathrm{m}$ working at $78\,\mathrm{dB}$.

(8) Calculate the sound pressure p_{horn} created by a $118\,\mathrm{dB}$ car horn.

(9) Give the values of the pain sound intensity I_{pain} and pain sound pressure p_{pain} and comment on the results of items (7) and (8) relative to pain values.

(10) Graduate work. Explain why different decibel-conversion formulas are required for sound intensity I and sound pressure p.

3 Extra Credit Challenge Problems

Challenge problems are optional.

C.1 Analyze Air Vibration in a Closed-Open Tube

(1) Sketch a tube closed at one end and open at the other end indicating all the information needed for the analysis.

(2) Write the boundary conditions at the two ends of the tube in terms of displacement $u(x,t)$ and pressure disturbance $p(x,t)$.

(3) Recall from HW01 the harmonic solution of the wave equation consisting of the multiplication of two terms:

- **(a)** First term: a function $U(x)$ consisting of trigonometric functions depending on wavenumber γ, space variable x, and constants C_1, C_2.
- **(b)** Second term: a complex exponential depending on frequency ω and time variable t.

(4) Recall from HW01 the expression of pressure $p(x,t)$ in terms of displacement $u(x,t)$ and bulk modulus K.

(5) Substitute (3) into (4) to obtain the pressure $p(x,t)$ in terms of constants C_1, C_2.

(6) Substitute (3) and (5) into (2) and write the boundary conditions at the two ends of the tube in terms of unknown constants C_1, C_2.

(7) Solve (6) for C_2 and obtain an equation for C_1.

(8) Discuss the trivial solution.

(9) Choose to search for the nontrivial solution and define the equation that needs to be solved to find a nontrivial solution.

(10) Introduce the notation $z = \gamma L$ into (9) and write the eigenvalue equation in terms of z.

(11) Solve the eigenvalue equation and write the general expression for the eigenvalues z_j, where $j = 1, 2, 3, \ldots$.

(12) Use (10) to write the general expression for the wavenumber γ_j in terms of j and length L.

(13) Recall from HW01 the definition of wavenumber γ in terms of frequency ω and wavespeed c.

(14) Solve (13) for frequency ω in terms of wavenumber γ and wavespeed c.

(15) Use (12) and (14) to write the general expression of the natural frequency ω_j in terms of j, wavespeed c, and tube length L.

(16) Use (15) to write the general expression of the Hz frequency f_j in terms of j, wavespeed c, and tube length L.

(17) Explain the meaning of fundamental frequency and overtones.

(18) Explain the meaning of fundamental harmonic and higher harmonics.

(19) Recall from HW01 the relationship between wavelength λ and wavenumber γ.

(20) Substitute (12) into (19) to write the general expression for the wavelength λ_j in terms of j and tube length L.

(21) Use (20) to explain which wavelength multiples could accommodate vibratory standing waves in a tube of length L closed at one end and open at the other end.

(22) Recall $U(x)$ from (3)(a) and use (7) to eliminate C_2.

(23) Set $C_1 = 1$ in item (22) and write the general expression for the modeshape $U_j(x)$ in terms of the wavenumber γ_j.

(24) Plot the normalized displacement modeshapes for modes 1 through 5 of air vibration in a tube of length L, and discuss your results.

(25) Graduate work. Calculate and plot the normalized pressure modeshapes for modes 1 through 5 of air vibration in a tube of length L, and discuss your results.

C.2 Analyze Air Vibration in an Open-Open Tube

(1) Sketch a tube open at both ends indicating the information needed for analysis.

(2) Write the boundary conditions at the two ends of the tube.

(3) Repeat all steps of Problem C.1 paying particular attention to the values taken by C_1 and C_2.

C.3 Calculate Air Vibration in a Tube with Various Boundary Conditions

Consider a tube of length $L = 0.85\,\text{m}$. Take sound speed $c = 340\,\text{m/s}$ and air density $\rho = 1.2\,\text{kg/m}^3$. The purpose is to calculate the modal vibration characteristics and modeshapes for $j = 1, \ldots, N$ with $N = 5$. The following boundary conditions are considered.

Case	$x = 0$	$x = L$
(i)	Closed	Open
(ii)	Open	Open

(1) Display the input data.

For each boundary-condition case, complete items (2)–(5).

(2) Calculate and display the modal vibration characteristics:

- **(a)** Eigenvalues z_j.
- **(b)** Wavenumbers γ_j.
- **(c)** Rad/s frequencies ω_j.
- **(d)** Hz frequencies f_j.

(3) Calculate and plot the displacement modeshapes.

(4) Compare your results with those of Problem B.4.

(5) Graduate work. Calculate and plot the normalized pressure modeshapes, and compare the result with item (3).

References

[1] National Aeronautics and Space Administration. Our high-energy universe: 15 years with the Fermi Gamma-ray Space Telescope. Technical Report SP-2024-4-254-GSFC, NASA's Goddard Space Flight Center, Greenbelt, Maryland, 2024. https://science.nasa.gov/wp-content/uploads/2024/04/our-high-energy-universe.pdf.

[2] The Nobel Prize. Waves or particles? Einstein explains the photoelectric effect. Nobel Prize Educational. Accessed October 16, 2026. https://educationalgames.nobelprize.org/educational/physics/quantised_world/waves-particles-2.html.

[3] Eite Tiesinga, Peter J. Mohr, David B. Newell, and Barry N. Taylor. CODATA recommended values of the fundamental physical constants: 2018. *Journal of Physical and Chemical Reference Data*, 50(3):033105, 2021. https://physics.nist.gov/cuu/pdf/JPCRD2018CODATA.pdf.